

 Norwegian University of Science and Technology NTNU Dep. of Electric Energy Address: 7491 Trondheim Contact: Noah Monz noah.l.monz@ntnu.no	TET4185 Power Market Spring 2026	
	Course project	
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This is the project assignment for TET4185.

The project consists of a report and a presentation to test your understanding. The project report may be a **maximum of 35**, A4 pages (excluding front and back matter). The deadline to submit the project report is the 5th of May.

Your group will give a 5-minute oral presentation of their report to the course coordinator and scientific assistant. After that, you and your group will also answer questions about the underlying theory from the course, including lectures and assignments, to demonstrate your theoretical understanding of the project tasks. More detailed information on the presentation will be given on Blackboard.

The group's final grade is determined by the project report (80 %) and the project presentation (20 %), based on the percentage of tasks completed in the project description and presentation. The final grading will be from A to F.

Grading rubric for the project:

- Problem 1: 15%
- Problem 2: 25%
- Problem 3: 20%
- Problem 4: 20%
- Oral project presentation: 20%

Gurobi is the preferred solver of the course. If you are unable to use Gurobi for whatever reason, GLPK can be used. Please state which solver is used. Please do not solve with both for analysis.

Problem 1: Introduction to OPF (15% of the project grade)

Problem 1 tests your knowledge of the fundamental theory and concepts of DCOPF and ATC in the context of a three-node system (which will also be looked at in Problem 2).

Calculations must be done with equations and without the use of extra tools, meaning that you must show the steps of your calculation.

A remote area somewhere in the Nordics consists of three villages, each with a load and a generation. The villages are connected through power lines. The layout of the system can be described as a three-node system, as shown in Figure 1.

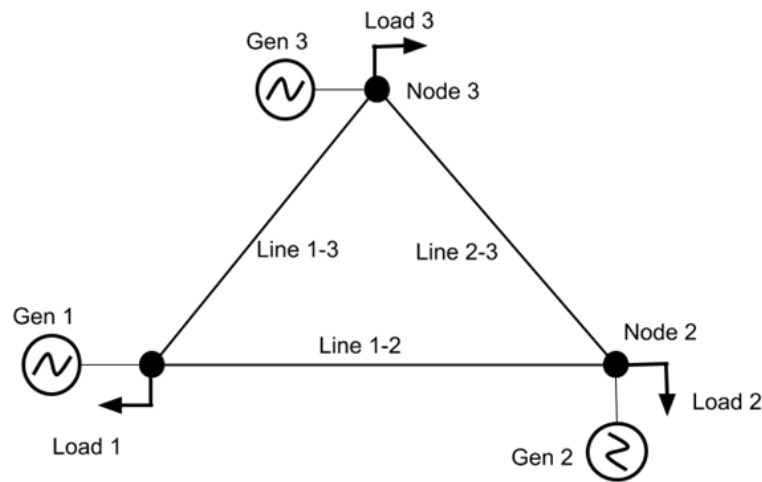


Figure 1: The three-node system

The transmission lines have some resistance that affects the power flow. This is reflected through their susceptance values b_{ij} . A small reminder, the susceptance is the inverse of the reactance in a line with no resistive losses. This is an approximation made for this type of Optimal Power Flow method.

For this scenario, we use the per-unit values based on a base value of 100 MVA.

Information about each node is the following:

Node	p_{gen} [NOK/MWh]	P_{cap} [MW]	P_{Load} [MW]
1	300	1000	200
2	1000	1000	200
3	600	1000	500

Information about each transmission line is the following:

Transmission line	Line Susceptance Magnitude [p.u] acquired from the Y- matrix
Line 1-2	20
Line 1-3	10
Line 2-3	30

Note: The admittances given in the table are the imaginary values $B, Y = G + jB$, meaning that these are the susceptances. Please note that to get the minus sign, the values should be put into the Y-matrix and read from there.

The following equations might be of help when analyzing using DCOPF:

Power equality in a node n with transmission lines L: $P_n = \sum_{l=1}^L b_{nl} \cdot (\delta_n - \delta_l)$

Power flow on a line: $P_{ij} = b_{ij} \cdot (\delta_i - \delta_j)$, note that P_{ij} must be given in per unit.

Task 1-1: Method approximation

This given AC grid problem can be solved using some method approximations. Two of these methods are the following:

- DC optimal power flow
- Transport Network (ATC)

Explain what both methods do, the considerations taking into account, and the important differences between them.

Task 1-2: Lagrangian expression of DCOPE

To help solve the DCOPF when there are multiple constraints present, the Lagrangian function is a helpful technique. A definition of this expression is:

"The Lagrangian function is a technique that combines the function being optimized with functions describing the constraint or constraints into a single equation. Solving the Lagrangian function allows you to optimize the variable you choose, subject to the constraints

you can't change.” [Source: <https://www.dummies.com/article/business-careers-money/business/economics/how-to-use-the-lagrangian-function-in-managerial-economics-166972/>]

Set up the Lagrangian expression for our current problem of the DCOPF, when we are not assuming any line or power production capacity constraints, and we want to minimize system cost.

Task 1-3: DCOPF without line constraint

In this task, we want to solve the given problem using the DC optimal power flow method. We let node 1 be considered a reference node (and we will do this for all following tasks), where the phase angle on this node is $\delta_1 = 0$.

Assume that the lines have no constraints on power flow, and find:

- Optimal generation
- Phase angles
- Power flows
- The prices in each node
- Total system cost

Task 1-4: DCOPF with line constraint

During a hot summer day, the transmission line is exposed to such heat that the capacity is reduced to uphold security constraints.

The new capacities of the lines are $P_{12,cap} = 500 \text{ MW}$, $P_{13,cap} = 500 \text{ MW}$, and $P_{23,cap} = 100 \text{ MW}$

- a) What happens when the lines are constrained in terms of production scheduling in each node? (no numerical answer needed, discussion only)
- b) Regarding the Lagrangian expression of DCOPF declared in Task 1-2, what additional constraint would we add to this regarding the line constraint?
- c) To solve the DCOPF, we must find the first-order condition* to find the expressions for OPF. Set this up, and define what variables are unknown. (You should end up with one new equation that enables you to find the shadow price for the line constraint)

- d) Solve this new problem and find using DCOPF:
- a. Optimal generation
 - b. Phase angles
 - c. Power Flows
 - d. The prices in each node
 - e. Total system cost
 - f. The shadow price of the constraint on Line 2-3.
- e) Provide reasoning for the nodal price results, particularly the nodal price at Node 2.
- f) If the demand at node 2 were increased to 201 MW, what would be the optimal generation and the total system cost? What is the change in system cost? Discuss your findings.

*: See “Notat Lagrangian” in the “Additional Material” folder on Blackboard, for an example of this.

Note: This theoretical solution is the basis for the second problem, in which you model this three-node system and use an optimizer to derive the optimal solution. Therefore, both problems 1 and 2 should arrive at the same solution.

Problem 2: Modeling a 3-node system (30% of the project grade)

Problem 2 asks you to write your own code to model a simple case of DCOPF. You are provided data and asked to formulate and solve the problem by writing your own model using Python and Pyomo.

Include in your submission the code as a “.py” file. Be sure to add comments in the code and use relevant variable names so the grader can understand the procedure.

For this problem, we have defined a 3-node system as shown in Figure 2. Numerical data on the system is attached in the Excel file “Problem 2 data.xlsx”, under the sheet “Problem 2.2 - Base case”. This problem will be solved using DCOPF and Flow-based market coupling.

Note: The admittances given in the Excel-file are the imaginary values $B, Y = G + jB$, meaning that these are the susceptances. Please note that to get the minus sign, the values should be put into the Y-matrix and read from there.

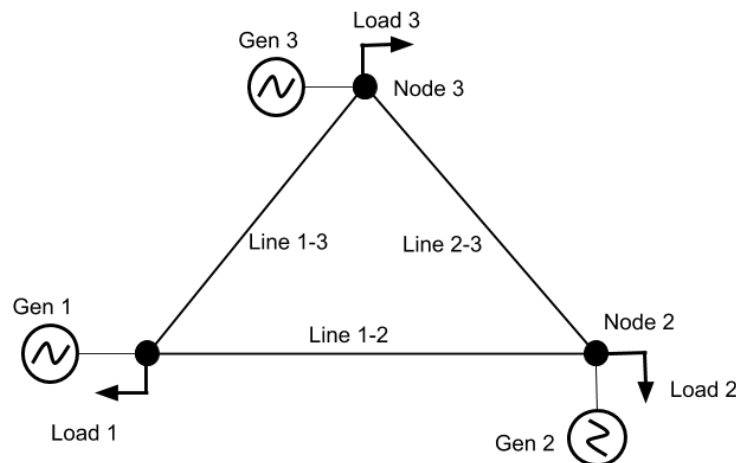


Figure 2: The 3-node system consisting of multiple generators and loads

Task 2-1: Defining the mathematical formulation for DCOPF

It is crucial to have a clear understanding of what is happening within the system that you try to model and to give it an accurate formulation that includes all the necessary considerations.

- Explain what the main objective of this problem formulation is. (What do we want to minimize or maximize?)
- What are the constraints regarding power generation?
- What are the constraints regarding power flow?
- Set up the total mathematical formulation for DCOPF.

Task 2-2: Creating the optimization problem

In this task, you are to create and solve the “Problem 2.2 - Base case” numerical example given in the attached Excel file, using optimization tools and programming skills. Set the model up in Python or Pyomo and show the results you obtain. You are free to choose how you set it up yourself. For this scenario, we use the per-unit values based on a base value of 1000 MVA.

- a) Solve the optimization problem for DCOPF. Analyze and present relevant results and discuss the system. What variables in the system are binding, and what are their corresponding shadow price?
- b) Set the generator's marginal cost at Node 3 to 1000 NOK/MWh and compute the updated nodal prices through analytical calculations. Explain the resulting nodal prices.
- c) Flow-based market coupling (FBMC) has been implemented in the Nordics Day-Ahead market. Briefly describe how flow-based modeling differs from the DCOPF approach. Compare the mathematical formulation of the FBMC and DCOPF approaches.
- d) Now, it is your turn to implement the FBMC approach and compare the results with the DCOPF approach. How do the results change? Explain your findings.

For all subsequent tasks in problem 2, you should only use the DCOPF method. You should not use the FBMC approach.

Task 2-3: Extending the model: Multiple generators

Node 1 has been extended to be made up of 3 different generators with different production capacities and marginal costs. Their data can be found in the sheet “Problem 2.3 - Generators”.

Extend the model so that all three generators are represented in Node 1 and solve the problem using DCOPF. Plot the production for each generator and find the nodal prices for each node. Discuss the results and relevant shadow prices.

Task 2-4: Extending the model: Multiple loads

Node 2 will now be extended to include 3 loads of different capacities. The data can be found in the sheet “Problem 2.4 - Loads”. First, consider only the capacity of the additional loads.

- a) Extend the model so that all three loads are represented in Node 2 and solve the problem using DCOPF. Plot the production for each generator and find the nodal prices for each node. Discuss the results and relevant shadow prices.

Now, also consider the marginal willingness-to-pay (WTP) for the additional loads at node 2. Considering the loads’ WTP allows us to **maximize social welfare** by considering flexibility from the consumer side. Assume the loads without any given WTP *must* be covered.

- b) Explain the change in the objective function required to modify the problem to social welfare maximization. Briefly explain and justify any assumptions made to ensure the inflexible loads are covered.
- c) Solve for maximum social welfare, including all generators and loads at each node. Discuss the results and relevant shadow prices.

Task 2-5: Extending the model: Environmental constraints

The data for this problem is found in “Problem 2.5 – Environmental”.

Every generating unit in the system has a greenhouse gas emission based on its production. After the Paris Agreement, regulators wish to reduce carbon dioxide emissions in the system. You are asked to assess alternative ways of doing this, building on your work from Task 2-4. One proposal calls for a Clean Energy Standard (CES) (a.k.a. green certificate scheme) to increase the adoption of zero-carbon energy resources. The proposal calls for a 20% CES across the entire 3-node system.

- A) Explain briefly (in a few sentences) how you would model this policy in the optimization model. What constraint will you add?
- B) Extend the model to include your new CES constraint. Analyze the results using DCOPF: how does the new system (i.e. the solution derived after solving the optimization problem) compare to the system before the implementation of the policy?
- C) What does the dual value of the new constraint represent? What are its units?

Another proposal is to implement a cap-and-trade (a.k.a. emissions trading system). You are asked to evaluate if this would be a better way to reduce emissions than the CES. To assess this, you are asked to test a cap-and-trade system that would achieve the same emissions across the 3-node system as the CES (i.e., the cap should limit emissions to a quantity that equals the system-wide emissions that resulted from when the CES was implemented, which you calculated in the previous step).

- a) Explain briefly (in a few sentences) how you would model this policy in the optimization model. What constraint will you add?
- b) Extend the model using DCOPF to include the cap and solve the optimization problem. Hereby, you can neglect the revenue flows from the trading part because you are only assessing the system as a whole and not the individual revenues of the producers. Therefore, you only have to consider the cap, which acts as an emission limit for the entire system. What does the dual value of the new constraint represent? What are its units?
- c) Analyze and compare the results between the two proposals (CES and Cap-and-trade) and the case without environmental constraints. Explore and discuss any differences in the technology mix, power prices, and transmission flows.
- d) Discuss the pros and cons of each policy. For example, is there a difference in cost-effectiveness between the two proposals?

Problem 3: Nordic System FBMC (20% of the project grade)

This problem asks you to analyze the behavior of power flow in the Nordic Region.

The Nordic Region is represented as a 12-node case representing bidding zones, and consists of Norway, Denmark, Sweden, and Finland (see Figure 3). Each bidding zone is simplified, resembling a single generator unit with a fixed generation cost and inelastic demand. All prices are given in Euros (€). **BE AWARE that this figure is for illustrative purposes only, and the data depicted are not the same as those in your assignment. Only use the data from the provided Excel spreadsheets.**

To investigate this 12-node case, you will use optimization tools. You are provided Python code for this problem.

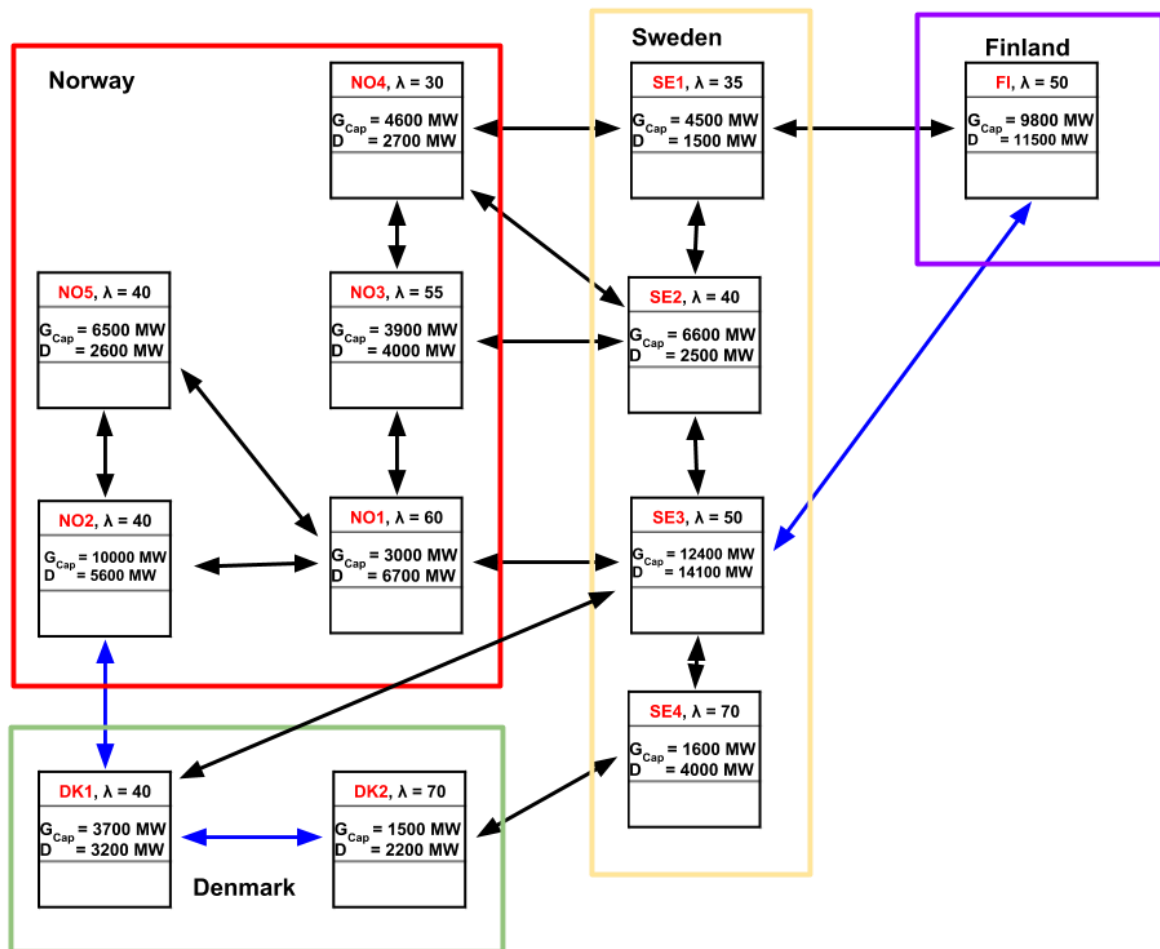


Figure 3: 12-node system for the Nordic System. The generation capacity is considered for an *exemplary* wet year (high hydropower production).

Each node has a generation capacity, load demand, and a nodal price for the generation of local power. Between each node are transmission lines. The **black colored lines** are AC

transmission lines, whereas **the blue lines** are HVDC cables. The HVDC cables are not limited in transmission capacity based on flow considerations, according to the Power Transfer Distribution Factors (PTDF), and can therefore freely choose power flow quantity while ignoring flow considerations according to PTDF.

The information regarding line capacity was found at NordPool, where we can find the NTC transfer capacities between the areas. The exact properties of the transmission lines are found in the Excel files attached. **Source:**

<https://www.nordpoolgroup.com/globalassets/download-center/tso/max-ntc.pdf>

This task will include two different versions with different generation capacities:

- a. A wet year, where the inflow has given the hydropower companies high production capacity. (Excel file: “Nordic_wet.xlsx”)
- b. A dry year, where the inflow has limited their production capacity. (Excel file: “Nordic_dry.xlsx”)

Unless specified in the task, you are to work on the wet-year version.

The model given is divided into the following (for the Python version):

- c. FBMC_pyomo.py -> The master script. This script takes in input data from a specified Excel file, stores the input data, creates an optimization problem that it solves, and stores the results in a separate Excel file. The output Excel file is given a specific name based on the method used (FBMC/ATC).
- d. Two Excel files that consist of the input data. The two files are similar except for the generation capacity (wet/dry year). The Excel files consist of node parameters such as demand, production capacity, and nodal prices, information regarding AC transmission lines, information regarding DC transmission lines, and a declaration of the optimization problem.

Task 3-1: Getting familiar with the model

- a) Take a look at the script for running the optimization model, and answer the following questions:
- What approximation methods does the model allow us to use to solve the 12-node system? How do we change which one to choose?
 - Where do we specify our reference node, and what does it mean?
 - What is the objective function of this problem? What are we trying to optimize?
 - If there is insufficient generation capacity to meet demand, what would the optimization model do?
 - What do you need to name the Excel file you are to analyze in order to run the script properly?
- b) Write down the mathematical formulation of both the FBMC and ATC problems.

NB: For the following tasks, we want you to present the relevant data using plots. Especially emphasize what you think are the most important observations.

Task 3-2: Analyzing a wet-year scenario

Attached to the file “Nordic_wet.xlsx” is the data for the base case scenario that you are to analyze. This base case considers a wet year, where the hydro reservoirs have high capacity due to high amounts of inflow over the year. Solve this scenario once **using the FBMC method** and once using the **ATC method**.

- a) Explain which approach is the more realistic depiction of the electricity system! How is this difference affecting the objective value, and why?
- b) Collect and compare data from the following:
- i. The objective function: total costs
 - ii. Node: Generation quantity and nodal prices
 - iii. Transmission: Power flow in all lines and the congestion rents.
- c) After FBMC was introduced in the Nordics, it has had a significant impact on market outcomes. In the lecture, you learned about the theoretical advantages of FBMC compared to the ATC approach. However, when you compare the social welfare solution obtained under the FBMC and ATC formulations in this simplified model, you

may observe that the results do not fully align with your expectations based on what you have learned in the lectures. **Explain why you see these particular results.** In your discussion, reflect on how modeling assumptions, network representation, and constraint structures may influence the outcomes.

Task 3-3: Dry-year scenario comparison

The file “Nordic_dry.xlsx” contains data for the same system in a scenario that is considered a dry year. This scenario has experienced a low amount of inflow to the hydro reservoirs, which limits their production quantity.

Solve this scenario for both **FBMC and ATC**, find the following, and compare the results you get:

- a) Node: Generation quantity and nodal prices
- b) Transmission: Power flow in all lines and the congestion rents.
- c) Compare the results from this task to the results obtained in Task 3-2 for the FBMC run. How do the nodal prices change when hydropower has a different contribution? How critical is it for Norway to have a good connection to other countries, depending on their seasonal experiences?

Task 3-4: Phasing out baseload production

Due to the high focus and controversy surrounding nuclear power production, the Swedish government decided to shut down all existing power plants in Sweden, which amounts to almost 8400 MW of power. All of the nodes are located in the node “SE3”. The data for the nuclear power plants are shown in the figure below:

Sweden's nuclear power reactors

Reactor	Operator ^b	Type	MWe net	Commercial operation	Intended decommissioning
Oskarshamn 3	OKG	BWR	1400	1985	2035 or 2045
Ringhals 1	Vattenfall	BWR	878	1976	2020
Ringhals 2	Vattenfall	PWR	807	1975	2019
Ringhals 3	Vattenfall	PWR	1062	1981	2041
Ringhals 4	Vattenfall	PWR	938	1983	2043
Forsmark 1	Vattenfall	BWR	984	1980	2040
Forsmark 2	Vattenfall	BWR	1120	1981	2041
Forsmark 3	Vattenfall	BWR	1187	1985	2045
Total (8)			8376		

Change the input data so that this power reduction of 8400 MW (we've rounded up from the table above) in "SE3" in Sweden is included, and run this scenario using FBMC (you do not need to run it for ATC). Compare this to the results you got in Task 3-2, and answer the following:

- What is the change in the objective function, and what is the reason behind this?
- How has the generation in each node changed, and where does the needed power come from now?
- How have the nodal prices changed? Who will be paying the most for this shift in production? Are there any areas that seem less affected by this change? How come?
- If Sweden were to do this in real life, what would be important to do beforehand to avoid having changes in power flow/production like those analyzed here?

Task 3-5: Environmental constraint:

The marginal costs for each node have previously excluded the carbon emissions from generation.

Go to <https://app.electricitymaps.com/map> and find the emission values for the relevant nodes in the system for a given hour (write down the day and hour of the data you want to use, the website updates regularly). For Sweden, assume the same emission value for all nodes.

Search online for prices for CO₂ emissions (also known as CO₂ quotas or European Union Allowances (EUAs)) for today, and possible future prices.

Find the mean CO₂ prices for the last X years and a CO₂ price forecast for the EU. With your new knowledge of zonal emissions and CO₂ prices, discuss:

- a) Which areas produce the most emissions? Is this explainable by their technology capacity share?
- b) What trend has been seen in changing CO₂ prices since 2005? What trend is forecasted until 2050?
- c) How do you expect the objective value and generation share to change with the inclusion of today's CO₂ price? What nodes are expected to behave differently? Which lines may experience more/less congestion rent?
- d) What trend do you expect to see in future prices, generation share, and total system cost in the future with the forecasted carbon prices?
- e) Implement a carbon price of 65€ in the code (only under FBMC). Are the results consistent with your assumptions?

Problem 4: Wind Integration and Price Spikes (20% of the project grade)

In this problem, you will use the Nordic flow-based market coupling (FBMC) model applied in Problem 3 to analyze how large shares of wind power affect dispatch, transmission congestion, and zonal electricity prices. **You will then construct stress scenarios that illustrate price spikes caused by:**

- 1. Peak demand**
- 2. Supply scarcity (low wind availability)**

Your analysis must explicitly link results to the merit-order effect, binding transmission constraints, scarcity pricing, and adequacy risks discussed in the lecture material.

4-1 Wind Integration in the Base (Wet-Year) Case

You are provided with an updated dataset where each bidding zone has the following additional wind capacity:

- WINDCAP (MW)
- WINDCOST = 0 €/MWh
- Assume GENMIN = 0 for wind

Nodes	NAMES	WINDCAP
1	"NO4"	500
2	"NO3"	800
3	"NO5"	500
4	"NO2"	1100
5	"NO1"	1300
6	"SE1"	400
7	"SE2"	600
8	"SE3"	4200
9	"SE4"	2000
10	"FI"	2300
11	"DK1"	3200
12	"DK2"	2200

You need to incorporate these wind capacities into the FBMC model (FBMC_pyomo.py).

Hereby, you may model wind as:

- A separate generator with zero marginal cost, or
- A reduction in net demand:

You must clearly state your modeling choice.

Run the **wet-year case** with wind **and compare it to the original wet-year case** (without wind from Problem 3-2).

Discuss:

1. Changes in zonal generation and prices
2. The merit-order effect from zero-cost wind
3. Changes in AC/HVDC flows
4. Which corridors become more/less congested
5. Congestion rents
6. Total system cost

Explain why zero-marginal-cost wind may reduce average prices but increase volatility.

4-2 Price Spikes from Peak Demand

Create **two peak-demand scenarios** by scaling **demand** upward in at least **two selected zones** (e.g., DK2 and SE4):

- Scenario A: +10% demand
- Scenario B: +20% demand

For each scenario, solve the FBMC model and compare zonal prices with the base wind case.

Identify:

1. Which constraints become binding?
2. Which marginal technology sets the price?
3. Why small demand increases can lead to disproportionately large price increases.

Relate your explanation to:

- Steep residual supply curves
- Binding transmission constraints
- Scarcity pricing mechanisms

4-3 Price Spikes from Supply Scarcity

Create a supply scarcity scenario by reducing wind availability:

Optionally combine this with reduced flexible generation (e.g., hydro capacity reduction) to simulate compounding scarcity.

Solve the scarcity scenario and compare with the base wind case. Now, discuss:

1. Changes in zonal prices
2. Congestion pattern shifts
3. Congestion rents
4. Which zones experience the largest price spikes, and why
5. Whether interconnectors (AC and HVDC) mitigate or amplify scarcity

Explain how low wind reduces the merit-order effect and exposes the system to marginal thermal pricing.